

AGRICULTURAL SYSTEMS TECHNOLOGY

Overview

The Department of Agricultural and Biosystems Engineering offers a Bachelor of Science degree in Agricultural Systems Technology (AST). Students majoring in AST choose between two options: Agricultural and Biosystems Management or Machine Systems. The department also offers a minor in Agricultural Systems Technology.

Successful AST graduates gain knowledge, skills, and abilities in solving technical problems, understanding the design process, excelling in authentic leadership, being aware of safety issues, having a quality orientation, effectively managing projects, and having a systems-thinking perspective. This translates to a holistic approach where AST graduates apply science, technology and engineering principles to manage complex agricultural and other production systems, including how the constituent sub-systems are interrelated and the broader impact of these systems.

Graduates find careers within a variety of agricultural industries, businesses, and organizations in the fields of agricultural machinery; off-road equipment; food, feed and grain processing; water quality; biorenewable resources; precision agriculture; or livestock production systems.

Common job duties of AST Agricultural and Biosystems Management graduates include:

- Technical support
- Operations management
- Production supervision
- Grain elevator management

Common job duties of AST Machine Systems graduates include:

- Precision agriculture support
- Testing or sales management
- Technical support
- Maintenance supervision

Student learning outcomes

Student Learning Outcomes

Upon graduation, all AST students should be able to:

1. Apply knowledge of mathematics, science, and applied engineering to identify and solve applied science and technology problems.
2. Develop and conduct experiments and analyze and interpret resulting data.

3. Evaluate and adapt systems, components, processes to meet specified needs.
4. Function effectively on multi-disciplinary teams.
5. Communicate effectively, ethically, and professionally in written, oral, and other formats to technical and non-technical audiences.
6. Understand the potential impacts and limitations of solutions in global and societal contexts.
7. Recognize the need for, and demonstrate an ability to, engage in life-long learning.
8. Effectively apply modern scientific and technical tools necessary for professional practice to address contemporary issues in applied engineering and technology.

Upon graduation, AST students in the agricultural and biosystems management (ABM) option should be able to:

1. Design, implement, and evaluate best practices for the management of global and natural resource systems.
2. Integrate and utilize agricultural and biosystems applied engineering and technology to address contemporary issues in bio-based industries.
3. Evaluate the factors impacting the complex systems that sustain water, air, soils, food, and feed.

Upon graduation, AST students in the machine systems (M.S.) option should be able to:

1. Specify, manage, and test machine systems in the context of a complete agricultural, biological production or processing system.
2. Use and apply the technology of machine systems including power and information flows, function and interaction with biological materials.
3. Perform an energy and cost analyses of complete machine systems to ensure the success and sustainability of an enterprise.

ACCREDITATION

For the undergraduate curriculum in Agricultural Systems Technology leading to a Bachelor of Science. The Agricultural Systems Technology program is accredited by the Association of Technology, Management, and Applied Engineering, using standards to accredit baccalaureate programs, <https://atmae.org/accreditation/>. ATMAE accredits degree programs in technology, technology management, applied technology, engineering technology, applied engineering, and technology-related disciplines delivered at nationally and regionally accredited institutions in the United States and institutions meeting the same standards internationally.

eXPLORE MORE

For more information about the AST degree: <http://www.abe.iastate.edu/undergraduate-students/agricultural-systems-technology> (<http://www.abe.iastate.edu/undergraduate-students/agricultural-systems-technology/>)

Contact Information

- **Primary Contact**
 - Jordan Smith
 - 1320 Elings Hall
 - [abe.iastate.edu](http://www.abe.iastate.edu) (<http://www.abe.iastate.edu/>)
 - josmith1@iastate.edu
 - 515-294-5196
- **Department Website** – Agricultural Systems Technology – Department of Agricultural and Biosystems Engineering (<http://www.abe.iastate.edu/undergraduate-students/agricultural-systems-technology/>)

Program Administration

- **Owning Academic Unit/College** – College of Agriculture and Life Sciences
- **Coordinating Academic Unit Department** – Agricultural and Biosystems Engineering - A Department

credential opportunities

- **Program Duration** – 4 years
- **Credential(s) Offered**
 - Agricultural Systems Technology, B.S. (120 credits)
- **Program Foci**
 - Agricultural & Biosystems Management Undergraduate Program Focus
 - Machine Systems Undergraduate Program Focus
- **Additional Major Note** – Agricultural Systems Technology is offered as an additional major.
- **Program Limitations** – None
- **Program Designation** – In-person

Total Degree Requirement: 120 cr.

Only 65 cr. from a two-year institution may apply which may include up to 16 technical cr.; 9 P-NP cr. of free electives; 2.00 minimum GPA.

Communications Proficiency:

6 cr. of English composition with a C or better and 3 cr. of speech fundamentals with a C or better.

Communication/Library: 13 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3

One of the following: 3

ENGL 3020	Business Communication	
ENGL 3090	Proposal and Report Writing	
ENGL 3140	Technical Communication	
AGEDS 3270	Survey of Agriculture and Life Sciences Communication	

One of the following: 3

SPCM 2120	Fundamentals of Public Speaking	
COMST 2140	Professional Communication	
AGEDS 3110	Presentation and Sales Strategies for Agricultural Audiences	

LIB 1600	Introduction to College Level Research	1
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Total Credits 13

Mathematical, Physical, and Life Sciences: 25 cr.

MATH 1450	Applied Trigonometry	3
MATH 1510	Calculus for Business and Social Sciences	3
STAT 1040	Introduction to Statistics	3
CHEM 1630	College Chemistry	4
CHEM 1630L	Laboratory in College Chemistry	1
PHYS 1310	General Physics I	4
PHYS 1310L	General Physics I Laboratory	1

One of the following: 3

BIOL 1010	Introductory Biology	
BIOL 2110	Principles of Biology I	
BIOL 2120	Principles of Biology II	
BIOL 2510	Biological Processes in the Environment	

Life Sciences Elective from approved College of Agriculture and Life Sciences list 3

Total Credits 25

Business, Humanities, Ethics, and Social Sciences: 18 cr.

ACCT 2840	Financial Accounting	3
or ACCT 2150	Legal Environment of Business	
ECON 1010	Principles of Microeconomics	3
TSM 3700	Occupational Safety (Ethics)	3
Humanities course from College of Agriculture and Life Sciences list		3
International Perspectives course from University list		3
U.S. Cultures and Communities course from University list		3

Total Credits 18

Technical Core: 27 cr.

TSM 1100	Introduction to Technology	1
TSM 1110	Experiencing Technology	1
TSM 1150	Solving Technology Problems	3

TSM 1160	Introduction to Design in Technology	3
TSM 2010	Preparing for Workplace Seminar	1
TSM 2100	Fundamentals of Technology	3
TSM 2140	Managing Technology Projects	1
TSM 3100	Total Quality Improvement	3
TSM 3631		4
TSM 3710	Occupational Safety Management	2
TSM 3970	Summer Internship in Technology	R
or TSM 3990	Internship in Technology	
TSM 4150	Applied Project Management in Technology	2
TSM 4160	Technology Capstone	3
Total Credits		27

TSM 3970 or 399 may count toward graduation

Agricultural and Biosystems Management Option: 37 cr.

TSM 4330	Precision Agriculture	3
Two of the following:		6
ECON 2300	Farm Business Management	
ECON 2350	Introduction to Agricultural Markets	
ECON 3320	Cooperatives	
ECON 3370	Agricultural Marketing	
MGMT 3700	Managing Organizations	
MGMT 3710	Organizational Behavior	
Select 20 credits from departmental list		20
8 credits of free electives		8
Total Credits		37

Machine Systems option: 37 cr.

TSM 2160	Advanced Technical Graphics, Interpretation, and CAD	2
ABE 2710, ABE 2720, or ABE 2730		1
TSM 2400	Introduction to Advanced Manufacturing: Metals and Plastics Processing	3
TSM 3300	Agricultural Machinery and Power Management	3
TSM 3350	Tractor Power	4
TSM 3370	Fluid Power Systems Technology	3
TSM 3650	Electrical Controls and Sensor Systems	3
TSM 4330	Precision Agriculture	3
TSM 4430	Statics and Strength of Materials for Technology	3
TSM 4650	Automation Systems	3
9 credits of free electives		9
Total Credits		37

Agricultural Systems Technology, B.S. - Machine Systems

First Year

Fall	Credits	Spring	Credits
TSM 1100	1	TSM 1110	1
TSM 1160	3	TSM 1150	3
ENGL 1500	3	MATH 1510	3
LIB 1600	1	PHYS 1310	4
MATH 1450	3	PHYS 1310L	1
CHEM 1630	4	U.S. Cultures and Communities - see list ¹	3
CHEM 1630L	1		
16		15	

Second Year

Fall	Credits	Spring	Credits
TSM 2010	1	TSM 2160	2
TSM 2100	3	ABE 2710, 2720, or 2730	1
TSM 2140	1	TSM 2400	3
TSM 3700	3	STAT 1040	3
ACCT 2840	3	BIOL 1010 or 2110	3
ENGL 2500	3	ECON 1010	3
14		15	

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
TSM 3350	4	TSM 3100	3	TSM 3970 or 3990	R
TSM 3631	4	TSM 3300	3		
TSM 4330	3	TSM 3370	3		
ENGL 3020, 3090, 3140, or AGEDS 3270	3	TSM 3650	3		
Elective	3	Life Science - See list ³	3		
17		15		0	

Fourth Year

Fall	Credits	Spring	Credits
TSM 4150		2 TSM 3710	2
TSM 4430		3 TSM 4160	3
SPCM 2120, COMST 2140, or AGEDS 3110		3 TSM 4650	3
International Perspectives - see list ¹		3 Humanities - See list ²	3
Elective		3 Elective	3
14		14	

¹ U.S. Cultures and Communities and International Perspectives (<https://www.registrar.iastate.edu/students/div-ip-guide/>)

² Humanities Course List (<https://www.cals.iastate.edu/student-services/humanities/>)

³ Life Sciences Course List (<https://www.cals.iastate.edu/student-services/life-science/>)

⁴ Ethics Course List (<https://www.cals.iastate.edu/student-services/ethics/>)

Agricultural Systems Technology, B.S. - Agricultural & Biosystems Management

First Year

Fall	Credits	Spring	Credits
TSM 1100		1 TSM 1110	1
TSM 1160		3 TSM 1150	3
ENGL 1500		3 MATH 1510	3
LIB 1600		1 PHYS 1310	4
MATH 1450		3 PHYS 1310L	1
CHEM 1630		4 ECON 1010	3
CHEM 1630L		1	
16		15	

Second Year

Fall	Credits	Spring	Credits
TSM 2010		1 BIOL 1010 or 2110	3
TSM 2100		3 STAT 1040	3

TSM 2140	1 SPCM 2120, COMST 2140, or AGEDS 3110	3
TSM 3700	3 ECON/ MGMT Elective	3
ACCT 2840 or 2150	3 Option Elective ⁵	4
ENGL 2500	3	
14		16

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
TSM 3631		4 TSM 3100		3 TSM 3970 or 3990	R
ENGL 3020, 3090, 3140, or AGEDS 3270		3 TSM 3710		2	
Option Elective ⁵		3 Option Elective ⁵		4	
Option Elective ⁵		3 Life Science - see list ³		3	
U.S. Cultures and Communities - see list ¹		3 Elective		2	
16		14		0	

Fourth Year

Fall	Credits	Spring	Credits
TSM 4150		2 TSM 4160	3
TSM 4330		3 TSM 3300	3
Humanities - see list ²		3 TSM 3650	3
Option Elective ⁵		3 International Perspectives - see list ¹	3
Option Elective ⁵		3 ECON/ MGMT Elective	3
14		15	

- ¹ U.S. Cultures and Communities and International Perspectives (<https://www.registrar.iastate.edu/students/div-ip-guide/>)
- ² Humanities Course List (<https://www.cals.iastate.edu/student-services/humanities/>)
- ³ Life Science Course List (<https://www.cals.iastate.edu/student-services/life-science/>)
- ⁴ Ethics Course List (<https://www.cals.iastate.edu/student-services/ethics/>)
- ⁵ Option Elective List (<https://www.abe.iastate.edu/undergraduate-students/agricultural-systems-technology/ast-curricula/>)

TSM 4710 Safety Laboratory

TSM 4770 Risk Analysis and Management

• At least six (6) credits of 3000-level or higher TSM classes (from the classes listed above)

• At least three (3) credits that are not used to meet any other department, college, or university requirement.

Minor in agricultural systems technology

The Department of Agricultural and Biosystems Engineering offers a minor in agricultural systems technology which may be earned by completing a minimum of 15 credits of technology systems management courses, which includes:

TSM 1150	Solving Technology Problems	3
TSM 2100	Fundamentals of Technology	3
9 credits from:		9
TSM 3100	Total Quality Improvement	
TSM 3220	Preservation of Grain Quality	
TSM 3220L	Preservation of Grain Quality Laboratory	
TSM 3240	Soil and Water Conservation Management	
TSM 3250	Biorenewable Systems	
TSM 3270	Livestock and Poultry Production Systems, Technology, and Management	
TSM 3300	Agricultural Machinery and Power Management	
TSM 3350	Tractor Power	
TSM 3370	Fluid Power Systems Technology	
TSM 3631	Electrical Power, Control, and Sensor Systems	
TSM 3650	Electrical Controls and Sensor Systems	
TSM 3710	Occupational Safety Management	
TSM 3720	Legal Aspects of Occupational Safety and Health	
TSM 3760	Fire Protection and Prevention	
TSM 3930E	Topics in Technology: Chemical Application Systems	
TSM 3930F	Topics in Technology: Agricultural Safety and Health	
TSM 4330	Precision Agriculture	
TSM 4550	Feed Processing and Technology	
TSM 4570	Feed Safety, Ingredient Quality and Analytics	
TSM 4650	Automation Systems	
TSM 4700	Basics of Industrial Hygiene	